

Questions with Answer Keys

#MathBoleTohMathonGo

Q1 - Derivative as Rate Measure

A 13ft ladder is leaning against a wall when its base starts to slide away. At the instant when the base is 12ft away from the wall, the base is moving away from the wall at the rate of 5ft/s. The rate at which the angle θ between the ladder and the ground is changing, is

(1) $-\frac{12}{13}\text{rad/s}$

(2) -1rad/s

(3) $-\frac{13}{12}\text{rad/s}$

(4) $\frac{10}{13}\text{rad/s}$

Q2 - Derivative as Rate Measure

Water is poured into an inverted conical vessel of which the radius of the base is 2 m and height 4 m at the rate of 77 L/min. The rate at which the water level is rising at the instant when the depth is 70 cm, is

(1) 10 cm/min

(2) 20 cm/min

(3) 40 cm/min

(4) None of these

Q3 - Derivative as Rate Measure

A point is moving along the curve $y^3 = 27x$. The interval in which the abscissa changes at slower rate than the ordinate, is

(1) $(-3, 3)$

(2) $(-\infty, \infty)$

(3) $(-1, 1)$

(4) $(-\infty, -3) \cup (3, \infty)$

Q4 - Derivative as Rate Measure

Questions with Answer Keys

#MathBoleTohMathonGo

A particle moves along the curve $y = x^{3/2}$ in the first quadrant in such a way that its distance from the origin increases at the rate of 11 units/s. The value of dx/dt when $x = 3$, is

- (1) 4
- (2) $9/2$
- (3) $3\sqrt{3}/2$
- (4) None of these

Q5 - Derivative as Rate Measure

If x and y are the sides of two squares such that $y = x - x^2$. Then, the rate of change, of the area of second square with respect to the first square, when $x = 2$, is

- (1) 1
- (2) 4
- (3) 3
- (4) 6

Q6 - Tangents and Normals

The equation of normal to the curve $y = x + \sin x \cos x$ at $x = \pi/2$ is

- (1) $x = 2$
- (2) $x = \pi$
- (3) $x + \pi = 0$
- (4) $2x = \pi$

Q7 - Tangents and Normals

The equation of normal to the curve $x + y = x^y$, where it cuts X -axis, is

- (1) $y = x + 1$
- (2) $y = -x + 1$
- (3) $y = x - 1$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(4) $y = -x - 1$

Q8 - Tangents and Normals

The number of values of c such that the straight line $3x + 4y = c$ touches the curve $\frac{x^4}{2} = x + y$, is

(1) 0

(2) 1

(3) 2

(4) 4

Q9 - Tangents and Normals

Let $f(x) = x^3 + ax + b$ with $a \neq b$ and suppose that tangent lines to the, graph of $f(x)$ at $x = a$ and $x = b$ have the same gradient. Then, the value of $f(1)$ is

(1) 0

(2) 1

(3) $-\frac{1}{3}$

(4) $\frac{2}{3}$

Q10 - Tangents and Normals

The coordinates of point(s) at each of which the tangents to the curve $y = x^3 - 3x^2 - 7x + 6$ cut off on the positive semi-axis OX a line segment half that on the negative semi-axis OY is given by

(1) $(-1, 9)$

(2) $(3, -15)$

(3) $(1, -3)$

(4) None of these

Q11 - Tangents and Normals

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

At the point $P(a, a^n)$ on the graph of $y = x^n (n \in \mathbb{N})$ in the first quadrant a normal is drawn. The normal intersects the Y -axis at the point $(0, b)$, if $\lim_{a \rightarrow 0} b = 1/2$, then n equals

- (1) 1
- (2) 2
- (3) 3
- (4) 4

Q12 - Tangents and Normals

If the tangent at P of the curve $y^2 = x^3$ intersects the curve again at Q and the straight lines OP, OQ make angles α, β with the X -axis, where O is the origin. Then, $\frac{\tan \alpha}{\tan \beta}$ has the value equal to $\tan \beta$

- (1) -1
- (2) -2
- (3) 2
- (4) $\sqrt{2}$

Q13 - Tangents and Normals

The equation of tangent to the curve $y = x + \frac{4}{x^2}$, which is parallel to the X -axis, is

- (1) $y = 0$
- (2) $y = 1$
- (3) $y = 2$
- (4) $y = 3$

Q14 - Tangents and Normals

The intercept on X -axis made by tangents to the curve, $y = \int_0^x |t| dt, x \in \mathbb{R}$, which are parallel to the line $y = 2x$, are equal to

- (1) ± 1
- (2) ± 2

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(3) ± 3

(4) ± 4

Q15 - Tangents and Normals

The tangent to the curve $y = e^x$ drawn at the point (c, e^c) intersects the line joining the points $(c-1, e^{c-1})$ and $(c+1, e^{c+1})$

(1) on the left of $x = c$ (2) on the right of $x = c$

(3) at no point

(4) at all points

Q16 - Mean Value Theorem

Consider the function for $x \in [-2, 3]$,

$$f(x) = \begin{cases} \frac{x^3 - 2x^2 - 5x + 6}{x-1}, & x \neq 1 \\ -6, & x = 1 \end{cases}$$

(1) f is discontinuous at $x = 1 \Rightarrow$ Rolle's is not applicable in $[-2, 3]$ (2) $f(-2) \neq f(3) \Rightarrow$ Rolle's theorem is not applicable in $[-2, 3]$ (3) f is not derivable in $(-2, 3) \Rightarrow$ Rolle's theorem is not applicable(4) Rolle's theorem is applicable as f satisfies all the conditions and c of Rolle's theorem is $1/2$

Q17 - Mean Value Theorem

If the function $f(x) = 2x^2 + 3x + 5$ satisfies LMVT at $x = 2$ on the closed interval $[1, a]$, then the value of a is

(1) 3

(2) 4

(3) 6

(4) 8

Questions with Answer Keys

#MathBoleTohMathonGo

Q18 - Mean Value Theorem

Consider the function $f(x) = 8x^2 - 7x + 5$ on the interval $[-6, 6]$. The value of c that satisfies the conclusion of the mean value theorem, is

- (1) $-7/8$
- (2) -4
- (3) $7/8$
- (4) 0

Q19 - Mean Value Theorem

Let f be a function which is continuous and differentiable for all real x . If $f(2) = -4$ and $f'(x) \geq 6$ for all $x \in [2, 4]$, then

- (1) $f(4) \leq 8$
- (2) $f(4) \geq 8$
- (3) $f(4) \geq 12$
- (4) None of these

Q20 - Mean Value Theorem

If $f(x) = x^3 + 7x - 1$, then $f(x)$ has a zero between $x = 0$ and $x = 1$. The theorem which best describes this, is

- (1) Rolle's theorem
- (2) Mean Value theorem
- (3) Maximum-minimum value theorem
- (4) Intermediate value theorem

Q21 - Mean Value Theorem

Consider $f(x) = |1 - x|$, $1 \leq x \leq 2$ and $g(x) = f(x) + b \sin\left(\frac{\pi}{2}x\right)$, $1 \leq x \leq 2$, then which of the following is correct?

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

- (1) Rolle's theorem is applicable to both f, g and $b = 3/2$
- (2) LMVT is not applicable to f and Rolle's theorem, if applicable to g with $b = 1/2$
- (3) LMVT is applicable to f and Rolle's theorem is applicable to g with $b = 1$
- (4) Rolle's theorem is not applicable to both f, g for any real b

Q22 - Mean Value Theorem

If $P(x) = 51x^{101} - 2323x^{100} - 45x + 1035$, then Rolle's is applicable for

- (1) $(45, 46)$
- (2) $(45, 46^{1/3})$
- (3) $(45^{1/100}, 46)$
- (4) $(45, 46^{1/100})$

Q23 - Monotonicity

The function $f(x) = x^2(x-2)^2$ is

- (1) increasing in $(0, 1) \cup (2, \infty)$
- (2) decreasing in $(0, 1) \cup (2, \infty)$
- (3) decreasing function
- (4) increasing function

Q24 - Monotonicity

The function $f(x) = x^3 + 6x^2 + (9 + 2k)x + 1$ is an increasing function, if

- (1) $k \geq 3/2$
- (2) $k > 3/2$
- (3) $k < 3/2$
- (4) $k \leq 3/2$

Q25 - Monotonicity

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

If $f(x) = \int \left(2 - \frac{1}{1+x^2} - \frac{1}{\sqrt{1+x^2}} \right) dx$, then f is

- (1) increasing in $(0, \infty)$ and decreasing in $(-\infty, 0)$
- (2) increasing in $(-\infty, 0)$ and decreasing in $(0, \infty)$
- (3) increasing in $(-\infty, \infty)$
- (4) decreasing in $(-\infty, \infty)$

Q26 - Monotonicity

For all $x \in (0, 1)$

- (1) $e^x < 1 + x$
- (2) $\log_{\theta}(1+x) < x$
- (3) $\sin x > x$
- (4) $\log_{\theta} x > x$

Q27 - Monotonicity

If $f(x) = xe^{x(1-x)}$, then $f(x)$ is

- (1) increasing on $[-1/2, 1]$
- (2) decreases on R
- (3) increasing on R
- (4) decreasing on $[-1/2, 1]$

Q28 - Monotonicity

If $f(x) > x, \forall x \in R$, then the equation $f[f(x)] - x = 0$ has

- (1) atleast one real root
- (2) more than one real root
- (3) no real root, if $f(x)$ is polynomial and a real root, if $f(x)$ is not a polynomial
- (4) no real root at all

Questions with Answer Keys

#MathBoleTohMathonGo

Q29 - Monotonicity

A function $y = f(x)$ is given by $x = \frac{1}{1+t^2}$ and $y = \frac{1}{t(1+t^2)}$ for all $t > 0$, then f is

- (1) increasing in $(0, 3/2)$ and decreasing in $(3/2, \infty)$
- (2) increasing in $(0, 1)$
- (3) increasing in $(0, \infty)$
- (4) None of the above

Q30 - Monotonicity

The function $f(x) = \frac{\log(\pi+x)}{\log(e+x)}$, is

- (1) increasing on $(0, \infty)$
- (2) decreasing on $(0, \infty)$
- (3) increasing on $(0, \pi/e)$, decreasing on $(\pi/e, \infty)$
- (4) None of the above

Q31 - Monotonicity

If $f(x) = a^{\{x\} \operatorname{Sgn} x}$, $g(x) = a^{[a^{|x|} \operatorname{sgn} x]}$ for $a > 0, a \neq 1$ and $x \in R$, where $\{\cdot\}$ and $[\cdot]$ denote the fractional part and greatest integer function respectively, then which of the following statements holds good for the function $h(x)$, where $(\log a)h(x) = \log(f(x)) + \log[g(x)]$?

- (1) h is even and increasing for $a > 1$
- (2) h is odd and decreasing for $a < 1$
- (3) h is even and decreasing for $a < 1$
- (4) h is odd and increasing for $a > 1$

Q32 - Monotonicity

Let $f(x) = x^3 + ax^2 + bx + 5 \sin^2 x$ be an increasing function in the set of real numbers R . Then, a and b satisfy the condition

- (1) $a^2 - 3b - 15 \geq 0$

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(2) $a^2 - 3b + 15 < 0$

(3) $a^2 - 3b - 15 < 0$

(4) $a > 0$ and $b > 0$

Q33 - Maxima Minima

If the function $f(x) = 2x^3 - 9ax^2 + 12a^2x + 1$, where $a > 0$, attains its maximum and minimum at p and q respectively such that $p^2 = q$, then a is

(1) $1/2$

(2) 3

(3) 1

(4) 2

Q34 - Maxima Minima

A triangular park is enclosed on two sides by a fence and on the third side by a straight river bank. The two sides having fence are of same length x . The maximum area enclosed by the park is

(1) $\sqrt{\frac{x^3}{8}}$

(2) $\frac{1}{2}x^2$

(3) πx^2

(4) $\frac{3}{2}x^2$

Q35 - Maxima Minima

Given $P(x) = x^4 + ax^3 + bx^2 + cx + d$ such that $x = 0$ is the only real root of $P'(x) = 0$. If $P(-1) < P(1)$, then in the interval $[-1, 1]$

(1) $P(-1)$ is the minimum but $P(1)$ is not the maximum at P (2) neither $P(-1)$ is the minimum nor $P(1)$ is the maximum at P (3) $P(-1)$ is the minimum and $P(1)$ is the maximum at P (4) $P(-1)$ is not the minimum but $P(1)$ is the maximum at P

Questions with Answer Keys

#MathBoleTohMathonGo

Q36 - Maxima Minima

The real number k for which the equation $2x^3 + 3x + k = 0$ has two distinct real roots in $[0, 1]$

- (1) lies between 1 and 2
- (2) lies between 2 and 3
- (3) lies between -1 and 0
- (4) does not exist

Q37 - Maxima Minima

A straight line L with negative slope passes through the point $(8, 2)$ and cuts the positive coordinates axes at points P and Q . Then, the absolute minimum of $OP + OQ$, as L varies (where O is origin), is

- (1) 15
- (2) 16
- (3) 17
- (4) 18

Q38 - Maxima Minima

Let f, g and h be real valued functions defined on the interval $[0, 1]$ by $f(x) = e^{x^2} + e^{-x^2}$, $g(x) = xe^{x^2} + e^{-x^2}$ and $h(x) = x^2e^{x^2} + e^{-x^2}$. If a, b and c denote respectively, the absolute maximum of f, g and h on $[0, 1]$, then

- (1) $a = b$ and $c \neq b$
- (2) $a = c$ and $a \neq b$
- (3) $a \neq b$ and $c \neq b$
- (4) $a = b = c$

Q39 - Maxima Minima

For all $a, b \in R$, the function $f(x) = 3x^4 - 4x^3 + 6x^2 + ax + b$ has

- (1) no extremum
- (2) exactly one extremum

#MathBoleTohMathonGo

Questions with Answer Keys

#MathBoleTohMathonGo

(3) exactly two extremum

(4) three extremum

Q40 - Maxima Minima

If $f(x) = \begin{cases} x, & 0 \leq x \leq 1 \\ 2 - e^{x-1}, & 1 < x \leq 2 \\ x - e, & 2 < x \leq 3 \end{cases}$ and $g(x) = \int_0^x f(t)dt, x \in [0, 3]$, then $g(x)$ has

(1) local maxima at $x = 1 + \log 2$ and local minima at $x = e$

(2) local maxima at $x = 1$ and local minima at $x = 2$

(3) no local maxima

(4) no local minima

Questions with Answer Keys

#MathBoleTohMathonGo

Answer Key

Q1 (2)	Q2 (2)	Q3 (3)	Q4 (1)
Q5 (3)	Q6 (4)	Q7 (3)	Q8 (2)
Q9 (2)	Q10 (2)	Q11 (2)	Q12 (2)
Q13 (4)	Q14 (1)	Q15 (1)	Q16 (4)
Q17 (1)	Q18 (4)	Q19 (2)	Q20 (2)
Q21 (3)	Q22 (3)	Q23 (1)	Q24 (1)
Q25 (3)	Q26 (2)	Q27 (1)	Q28 (4)
Q29 (2)	Q30 (2)	Q31 (4)	Q32 (2)
Q33 (4)	Q34 (2)	Q35 (4)	Q36 (4)
Q37 (4)	Q38 (4)	Q39 (2)	Q40 (1)
